VANISHING ABOVE THE NUMBER OF SHEETS OF AN AFFINE ETALE COVER OF A QUASI-COMPACT SEPARATED SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The alternating Cech spectral sequence attached to a surjective etale morphism. Same chapter spaces-cohomology.tex, label lemma-alternating-spectral-sequence (invoked at proof line 1345). It computes the cohomology of X from the twisted cohomology of the configuration quotients U_p, and requires f to be separated, which holds because U and X are separated.
- Separatedness of the composite, and the finite flat equivalence for groupoid quotients. Morphisms of Spaces, label spaces-morphisms-lemma-compose-after-separated (invoked at proof line 1348) and Groupoids, label groupoids-proposition-finite-flat-equivalence (invoked at proof line 1354). The first gives that f is separated; the second identifies $U_p = W_p/S_{p+1}$ as an affine scheme because W_p is affine and the action is free and finite.
- Vanishing of the higher cohomology of a quasi-coherent module on the affine configuration scheme, and quasi-coherence of the twists. Cohomology of Schemes, label coherent-lemma-quasi-coherent-affine-cohomology-zero (invoked at proof line 1361) and Same chapter spaces-cohomology.tex, label lemma-quasi-coherent-twist (invoked at proof line 1363). They make the E_1-terms vanish in positive degrees: the twisted sheaves are quasi-coherent, their cohomology on W_p agrees with the cohomology of the corresponding module on the affine scheme, and that vanishes above degree zero.

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Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Navigation pointer for \ref{coherent-lemma-quasi-coherent-affine-cohomology-zero} used inside the counted span (line 1361); the argument that consumes it is retained in the span.

Source reference. Original proposition 03BM: sources/stacks-chow/tags.txt, source lines 4353–4353. Navigation pointer for \ref{groupoids-proposition-finite-flat-equivalence} used inside the counted span (line 1354); the argument that consumes it is retained in the span.

Source reference. Original lemma 0728: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1217–1217. Navigation pointer for \ref{lemma-alternating-spectral-sequence} used inside the counted span (line 1345); the argument that consumes it is retained in the span.

Source reference. Original lemma 072A: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1305–1305. Navigation pointer for \ref{lemma-quasi-coherent-twist} used inside the counted span (line 1363); the argument that consumes it is retained in the span.

Source reference. Original section 071Y: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 288–288. Navigation pointer for \ref{section-higher-direct-image} used inside the counted span (line 1356); the argument that consumes it is retained in the span.

Source reference. Original lemma 03KR: sources/stacks-chow/tags.txt, source lines 4687–4687. Navigation pointer for \ref{spaces-morphisms-lemma-compose-after-separated} used inside the counted span (line 1348); the argument that consumes it is retained in the span.

Proposition 0.1. *Let S be a scheme. Let X be an algebraic space over S. Assume X is quasi-compact and separated. Let U be an affine scheme, and let $f : U \to X$ be a surjective étale morphism. Let d be an upper bound for the size of the fibres of $|U| \to |X|$. Then for any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $H^q(X, \mathcal{F}) = 0$ for $q \geq d$.*

Proof. We will use the spectral sequence of Lemma 0728. The lemma applies since f is separated as U is separated, see Morphisms of Spaces, Lemma 03KR. Since X is separated the scheme $U \times_X \ldots \times_X U$ is a closed subscheme of $U \times_{\mathrm{Spec}(\mathbf{Z})} \ldots \times_{\mathrm{Spec}(\mathbf{Z})} U$ hence is affine. Thus W_p is affine. Hence $U_p = W_p/S_{p+1}$ is an affine scheme by Groupoids, Proposition 03BM. The discussion in Section 071Y shows that cohomology of quasi-coherent sheaves on W_p (as an algebraic space) agrees with the cohomology of the corresponding quasi-coherent sheaf on the underlying affine scheme, hence vanishes in positive degrees by Cohomology of Schemes, Lemma 01XB. By Lemma 072A the sheaves $\mathcal{F}|_{U_p} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi_p)$ are quasi-coherent. Hence $H^q(W_p, \mathcal{F}|_{U_p} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi_p))$ is zero when $q > 0$. By our definition of the integer d we see that $W_p = \emptyset$ for $p \geq d$. Hence also $H^0(W_p, \mathcal{F}|_{U_p} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi_p))$ is zero when $p \geq d$. This proves the proposition. □